- *Human-in-the-loop Learning*: This involves continuous interaction and feedback from human users or experts to guide the model's learning process. For instance, a legal LLM could be continuously updated based on feedback from legal professionals using the model. This feedback could be incorporated in the form of additional training data, changes to the model's reward function in a reinforcement learning framework, or modifications to the model's prompts. This could lead to a more dynamic and adaptable model that can evolve with the needs and knowledge of the users.
- *Active Learning*: This approach involves the model actively querying for information or feedback when it encounters a domain-specific concept it doesn't understand or has low confidence about. For instance, if a model trained on general news articles encounters a specialized medical term it doesn't understand, it could query a medical ontology or ask for clarification from a human user. The model could then incorporate this new information into its subsequent responses. This could make the model more effective at handling unfamiliar domain-specific topics, and could provide a more interactive and engaging user experience.

Each of these techniques provides a promising direction for future research in the domain specialization of large language models, and could help address some of the challenges and limitations of the current black-box, grey-box, and white-box methods.

8 CONCLUSION

In conclusion, the rapid advancement of LLMs has sparked significant interest in harnessing their potential to tackle domain-specific tasks in various natural, social, and formal science fields. However, several challenges, such as limited domain-specific expertise, knowledge elicitation, and model complexity, hinder the direct application of LLMs in these domains. This survey systematically categorizes and summarizes existing domain specialization techniques based on their access level to LLMs, along with a comprehensive overview of application domains that can benefit from specialized LLMs. By offering a detailed analysis of the advantages, disadvantages, and relationships among different techniques and domains, this survey aims to assist domain experts in identifying suitable techniques for their target problem settings, while also providing data scientists with a clear understanding of the practical significance and open challenges in various application domains. Moreover, the paper highlights the current status of research in this area, shedding light on future trends and potential avenues for interdisciplinary collaboration. As the field of LLM domain specialization continues to evolve, this survey serves as a valuable resource for researchers and practitioners, fostering further advancements and innovations in the application of artificial intelligence across diverse domains.

REFERENCES

[1] Zeyuan Allen-Zhu, Yuanzhi Li, and Zhao Song. 2019. A convergence theory for deep learning via over-parameterization. In *International Conference on Machine Learning*. PMLR, 242–252.

[2] Simran Arora, Avanika Narayan, Mayee F Chen, Laurel Orr, Neel Guha, Kush Bhatia, Ines Chami, and Christopher Re. 2023. Ask Me Anything: A simple strategy for prompting language models. In *The Eleventh International Conference on Learning Representations*.

[3] Akari Asai, Mohammadreza Salehi, Matthew E Peters, and Hannaneh Hajishirzi. 2022. Attempt: Parameter-efficient multi-task tuning via attentional mixtures of soft prompts. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. 6655–6672.

[4] Zhangir Azerbayev, Ansong Ni, Hailey Schoelkopf, and Dragomir Radev. 2022. Explicit Knowledge Transfer for Weakly-Supervised Code Generation. *arXiv preprint arXiv:2211.16740* (2022).

[5] Yejin Bang, Samuel Cahyawijaya, Nayeon Lee, Wenliang Dai, Dan Su, Bryan Wilie, Holy Lovenia, Ziwei Ji, Tiezheng Yu, Willy Chung, et al. 2023. A multitask, multilingual, multimodal evaluation of chatgpt on reasoning, hallucination, and interactivity. *arXiv preprint arXiv:2302.04023* (2023).

[6] Ankur Bapna, Naveen Arivazhagan, and Orhan Firat. 2019. Simple, scalable adaptation for neural machine translation. *arXiv preprint arXiv:1909.08478* (2019).

[7] Eyal Ben-David, Nadav Oved, and Roi Reichart. 2022. PADA: Example-based Prompt Learning for on-the-fly Adaptation to Unseen Domains. *Transactions of the Association for Computational Linguistics* 10 (2022), 414–433.

[8] Rishabh Bhardwaj, Amrita Saha, and Steven C. H. Hoi. 2022. Vector-Quantized Input-Contextualized Soft Prompts for Natural Language Understanding. In *Conference on Empirical Methods in Natural Language Processing*.

[9] Som S Biswas. 2023. Potential Use of Chat GPT in Global Warming. *Annals of Biomedical Engineering* (2023), 1–2.

[10] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George Bm Van Den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, et al. 2022. Improving language models by retrieving from trillions of tokens. In *International conference on machine learning*. 2206–2240.

[11] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems* 33 (2020), 1877–1901.

[12] Lucas Caccia, Edoardo Ponti, Lucas Liu, Matheus Pereira, Nicolas Le Roux, and Alessandro Sordoni. 2022. Multi-Head Adapter Routing for Data-Efficient Fine-Tuning. *arXiv preprint arXiv:2211.03831* (2022).

[13] Yihan Cao, Siyu Li, Yixin Liu, Zhiling Yan, Yutong Dai, Philip S Yu, and Lichao Sun. 2023. A comprehensive survey of ai-generated content (aigc): A history of generative ai from gan to chatgpt. *arXiv preprint arXiv:2303.04226* (2023).

[14] Yekun Chai, Shuohuan Wang, Yu Sun, Hao Tian, Hua Wu, and Haifeng Wang. 2022. Clip-Tuning: Towards Derivative-free Prompt Learning with a Mixture of Rewards. In *Conference on Empirical Methods in Natural Language Processing*.

[15] Ilias Chalkidis, Manos Fergadiotis, Prodromos Malakasiotis, Nikolaos Aletras, and Ion Androutsopoulos. 2020. LEGAL-BERT: The Muppets straight out of Law School. In *Findings of the Association for Computational Linguistics: EMNLP 2020*. 2898–2904.

[16] Sanyuan Chen, Yutai Hou, Yiming Cui, Wanxiang Che, Ting Liu, and Xiangzhan Yu. 2020. Recall and learn: Fine-tuning deep pretrained language models with less forgetting. *arXiv preprint arXiv:2004.12651* (2020).

[17] Wenhu Chen, Xueguang Ma, Xinyi Wang, and William W. Cohen. 2022. Program of Thoughts Prompting: Disentangling Computation from Reasoning for Numerical Reasoning Tasks. *arXiv preprint arXiv:2211.12588* (2022).

[18] Xiang Chen, Ningyu Zhang, Ningyu Zhang, Xin Xie, Shumin Deng, Yunzhi Yao, Chuanqi Tan, Fei Huang, Luo Si, and Huajun Chen. 2021. KnowPrompt: Knowledge-aware Prompt-tuning with Synergistic Optimization for Relation Extraction. *Proceedings of the ACM Web Conference 2022* (2021).

[19] Zhoujun Cheng, Tianbao Xie, Peng Shi, Chengzu Li, Rahul Nadkarni, Yushi Hu, Caiming Xiong, Dragomir Radev, Mari Ostendorf, Luke Zettlemoyer, Noah A. Smith, and Tao Yu. 2023. Binding Language Models in Symbolic Languages. In *The Eleventh International Conference on Learning Representations*.

[20] Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, et al. 2022. Palm: Scaling language modeling with pathways. *arXiv preprint arXiv:2204.02311* (2022).

[21] Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. Deep reinforcement learning from human preferences. *Advances in neural information processing systems* 30 (2017).

[22] Alexandra Chronopoulou, Matthew E Peters, Alexander Fraser, and Jesse Dodge. 2023. AdapterSoup: Weight Averaging to Improve Generalization of Pretrained Language Models. *arXiv preprint arXiv:2302.07027* (2023).

[23] Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Eric Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, et al. 2022. Scaling instruction-finetuned language models. *arXiv preprint arXiv:2210.11416* (2022).

[24] Hejie Cui, Jiaying Lu, Shiyu Wang, Ran Xu, Wenjing Ma, Shaojun Yu, Yue Yu, Xuan Kan, Chen Ling, Joyce Ho, et al. 2023. A Survey on Knowledge Graphs for Healthcare: Resources, Applications, and Promises. *arXiv preprint arXiv:2306.04802* (2023).

[25] Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. 2021. Knowledge neurons in pretrained transformers. *arXiv preprint arXiv:2104.08696* (2021).

[26] Zhuyun Dai, Vincent Y Zhao, Ji Ma, Yi Luan, Jianmo Ni, Jing Lu, Anton Bakalov, Kelvin Guu, Keith B Hall, and Ming-Wei Chang. 2022. Promptagator: Few-shot dense retrieval from 8 examples. *arXiv preprint arXiv:2209.11755* (2022).

[27] Ishita Dasgupta, Christine Kaeser-Chen, Kenneth Marino, Arun Ahuja, Sheila Babayan, Felix Hill, and Rob Fergus. 2023. Collaborating with language models for embodied reasoning. In *Second Workshop on Language and Reinforcement Learning*.

[28] Nicola De Cao, Wilker Aziz, and Ivan Titov. 2021. Editing Factual Knowledge in Language Models. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*.

[29] Mingkai Deng, Jianyu Wang, Cheng-Ping Hsieh, Yihan Wang, Han Guo, Tianmin Shu, Meng Song, Eric P. Xing, and Zhiting Hu. 2022. RLPrompt: Optimizing Discrete Text Prompts with Reinforcement Learning. In *Conference on Empirical Methods in Natural Language Processing*.

[30] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* (2018).

[31] Ning Ding, Yujia Qin, Guang Yang, Fuchao Wei, Zonghan Yang, Yusheng Su, Shengding Hu, Yulin Chen, Chi-Min Chan, Weize Chen, et al. 2022. Delta tuning: A comprehensive study of parameter efficient methods for pre-trained language models. *arXiv* (2022).

[32] Andrew Drozdov, Nathanael Schärli, Ekin Akyürek, Nathan Scales, Xinying Song, Xinyun Chen, Olivier Bousquet, and Denny Zhou. 2023. Compositional Semantic Parsing with Large Language Models. In *The Eleventh International Conference on Learning Representations*.

[33] Dheeru Dua, Shivanshu Gupta, Sameer Singh, and Matt Gardner. 2022. Successive Prompting for Decomposing Complex Questions. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*.

[34] Ali Edalati, Marzieh Tahaei, Ivan Kobyzev, Vahid Partovi Nia, James J Clark, and Mehdi Rezagholizadeh. 2022. KronA: Parameter Efficient Tuning with Kronecker Adapter. *arXiv preprint arXiv:2212.10650* (2022).

[35] Mohammad Fahes, Tuan-Hung Vu, Andrei Bursuc, Patrick Pérez, and Raoul de Charette. 2022. P{\O}DA: Prompt-driven Zero-shot Domain Adaptation. *arXiv preprint arXiv:2212.03241* (2022).

[36] Yang Feng, Shiyue Zhang, Andi Zhang, Dong Wang, and Andrew Abel. 2017. Memory-augmented Neural Machine Translation. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*.

[37] Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. 2022. PAL: Program-aided Language Models. *arXiv preprint arXiv:2211.10435* (2022).

[38] Chunjiang Ge, Rui Huang, Mixue Xie, Zihang Lai, Shiji Song, Shuang Li, and Gao Huang. 2022. Domain Adaptation via Prompt Learning. *arXiv preprint arXiv:2202.06687* (2022).

[39] Edouard Grave, Armand Joulin, and Nicolas Usunier. 2017. Improving neural language models with a continuous cache. In *International Conference on Learning Representations*.

[40] Yuxian Gu, Xu Han, Zhiyuan Liu, and Minlie Huang. 2022. PPT: Pre-trained Prompt Tuning for Few-shot Learning. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Dublin, Ireland, 8410–8423.

[41] Xu Guo, Boyang Li, and Han Yu. 2022. Improving the Sample Efficiency of Prompt Tuning with Domain Adaptation. In *Findings of the Association for Computational Linguistics: EMNLP 2022*. Association for Computational Linguistics, Abu Dhabi, United Arab Emirates.

[42] Xu Guo and Han Yu. 2022. On the Domain Adaptation and Generalization of Pretrained Language Models: A Survey. *arXiv* (2022).

[43] Karen Hambardzumyan, Hrant Khachatrian, and Jonathan May. 2021. WARP: Word-level Adversarial ReProgramming. In *Annual Meeting of the Association for Computational Linguistics*.

[44] Hangfeng He, Hongming Zhang, and Dan Roth. 2022. Rethinking with Retrieval: Faithful Large Language Model Inference. *arXiv preprint arXiv:2301.00303* (2022).

[45] Junxian He, Chunting Zhou, Xuezhe Ma, Taylor Berg-Kirkpatrick, and Graham Neubig. 2021. Towards a unified view of parameter-efficient transfer learning. *arXiv preprint arXiv:2110.04366* (2021).

[46] Shwai He, Liang Ding, Daize Dong, Miao Zhang, and Dacheng Tao. 2022. Sparseadapter: An easy approach for improving the parameter-efficiency of adapters. *arXiv preprint arXiv:2210.04284* (2022).

[47] Dan Hendrycks and Kevin Gimpel. 2016. Gaussian error linear units (gelus). *arXiv preprint arXiv:1606.08415* (2016).

[48] Evan Hernandez, Belinda Z Li, and Jacob Andreas. 2023. Measuring and Manipulating Knowledge Representations in Language Models. *arXiv preprint arXiv:2304.00740* (2023).

[49] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. 2019. Parameter-efficient transfer learning for NLP. In *International Conference on Machine Learning*. PMLR, 2790–2799.

[50] Jeremy Howard and Sebastian Ruder. 2018. Universal language model fine-tuning for text classification. *arXiv preprint arXiv:1801.06146* (2018).

[51] Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alexander Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. 2023. Distilling Step-by-Step! Outperforming Larger Language Models with Less Training Data and Smaller Model Sizes. *arXiv preprint arXiv:2305.02301* (2023).

[52] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685* (2021).

[53] Shengding Hu, Ning Ding, Huadong Wang, Zhiyuan Liu, Jingang Wang, Juanzi Li, Wei Wu, and Maosong Sun. 2022. Knowledgeable Prompt-tuning: Incorporating Knowledge into Prompt Verbalizer for Text Classification. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Dublin, Ireland.

[54] Zhiqiang Hu, Yihuai Lan, Lei Wang, Wanyu Xu, Ee-Peng Lim, Roy Ka-Wei Lee, Lidong Bing, and Soujanya Poria. 2023. LLM-Adapters: An Adapter Family for Parameter-Efficient Fine-Tuning of Large Language Models. *arXiv preprint arXiv:2304.01933* (2023).

[55] Jiaxin Huang, Shixiang Shane Gu, Le Hou, Yuexin Wu, Xuezhi Wang, Hongkun Yu, and Jiawei Han. 2022. Large language models can self-improve. *arXiv preprint arXiv:2210.11610* (2022).

[56] Shaohan Huang, Li Dong, Wenhui Wang, Yaru Hao, Saksham Singhal, Shuming Ma, Tengchao Lv, Lei Cui, Owais Khan Mohammed, Qiang Liu, et al. 2023. Language is not all you need: Aligning perception with language models. *arXiv preprint arXiv:2302.14045* (2023).

[57] Gautier Izacard, Patrick Lewis, Maria Lomeli, Lucas Hosseini, Fabio Petroni, Timo Schick, Jane Dwivedi-Yu, Armand Joulin, Sebastian Riedel, and Edouard Grave. 2022. Few-shot learning with retrieval augmented language models. *arXiv preprint arXiv:2208.03299* (2022).

[58] Katharina Jeblick, Balthasar Schachtner, Jakob Dexl, Andreas Mittermeier, Anna Theresa Stüber, Johanna Topalis, Tobias Weber, Philipp Wesp, Bastian Sabel, Jens Ricke, et al. 2022. ChatGPT Makes Medicine Easy to Swallow: An Exploratory Case Study on Simplified Radiology Reports. *arXiv preprint arXiv:2212.14882* (2022).

[59] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *Comput. Surveys* 55, 12 (2023), 1–38.

[60] Chen Jia and Yue Zhang. 2022. Prompt-based Distribution Alignment for Domain Generalization in Text Classification. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. 10147–10157.